

The Indefinite Block in the Unified Elastic Solve

Newton Physics Engine — reduced-elastic-sparse-tests

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What was wrong

A reduced elastic body is a floating frame (6 DOF) plus n_m modal coordinates. Commit 06275246 merged the two into one $(6 + n_m)$ dense block. Positive definiteness of that block requires its three partitions $\mathbf{H}_{ff}$, $\mathbf{H}_{fm}$, $\mathbf{H}_{mm}$ to come from a single $\mathbf{J}^\top \mathbf{K} \mathbf{J}$. They did not.

Each elastic contact was evaluated twice, on different geometry.

	$\mathbf{H}_{ff}$ (frame)	$\mathbf{H}_{fm}$, $\mathbf{H}_{mm}$ (cross, modal)
evaluator	<code>...from_collision</code>	<code>...from_world_points</code>
contact point	local point $\times$ rigid body_q	deformed, sample + $\sum_i q_i \phi_i$
levers	split (skeleton / anchor)	one lever
damping gate	rigid approach rate	deformed approach rate
friction slip	rigid anchor	deformed sample

The caller computed the deformed point, used it for the scalar penetration, and then passed the *local* point to the frame evaluator. Everything entering $\mathbf{H}_{ff}$ was therefore computed on the undeformed blade.

Fitting a contact stiffness $\mathbf{K} = a(\mathbf{I} - \mathbf{nn}^\top) + b\mathbf{nn}^\top$ separately from each side over 62 blocks gives $a_{\text{elastic}}/a_{\text{frame}}$ between 0.4 and 14, median 0.6 — not a constant, so not a units or duplication bug. The resulting deficiency is two-dimensional in 90% of indefinite blocks and lies in the friction tangent plane (overlap with the contact normal 0.000–0.021).

Consequences: $\lambda_{\min}$ down to -6.5×10^8 ; `wp.tile_cholesky` returns a finite but meaningless factor; the only guard is an `isfinite` check that never fires; the applied step has $\|\mathbf{H}\boldsymbol{\delta} + \mathbf{g}\|/\|\mathbf{g}\| \approx 1$ instead of 10^{-6} . Both rigid solve paths were affected.

The sparse path added a second, independent defect. It solved the full articulation including the blade and then *discarded* the elastic delta without pinning the block first, so applying only $\boldsymbol{\delta}_r$ left

$$\mathbf{g}_r^+ = (1 - \omega) \mathbf{g}_r - \omega \mathbf{H}_{re} \boldsymbol{\delta}_e, \quad \|\mathbf{H}_{re}\| \sim k = 10^6,$$

a spurious force on every neighbour each iteration. That is why the sparse path reached 10^{15} while the local path, running the same broken block, stayed bounded.

Why it surfaced only after the port. Both evaluators, and the inconsistency between them, are byte-identical before and after 06275246; the port changed neither. Before the merge the two results went into *separate* matrices, each individually positive definite, so the disagreement was a block Gauss–Seidel splitting error that cost convergence rate and nothing else. The merge put them in one matrix and silently promoted it into a violated definiteness precondition. This is unique to this branch: the upstream elastic branches have no AVBD contact — no dual, no hard contacts, no stabilization — so both call sites go through the deformed-point evaluator and `...from_collision` is never called at all.

What the changes do

1. One evaluation, three projections (`rigid_vbd_kernels.py`). A new `evaluate_contact_force_hessian_block` dispatches on whether the contact is elastic:

```

if is_elastic:
    return evaluate_rigid_contact_from_world_points(...) # deformed points
return evaluate_rigid_contact_from_collision(...)        # rigid, unchanged

```

Both frame call sites (`accumulate_body_body_contacts_per_body`, `compute_rigid_contact_forces`) now go through it, passing the deformed points they already computed and previously discarded. An elastic contact is therefore evaluated by the same function with the same arguments on the frame side and the modal side, so the block is a partition of one $\mathbf{J}^\top \mathbf{K} \mathbf{J}$ by construction. Rigid contacts keep `from_collision` untouched, with AVBD’s split levers and duals.

2. The frame is excluded from the sparse system (`rigid_sparse_articulation_kernels.py`). The mechanism already existed for kinematic bodies; elastic bodies now use it:

```

if body_inv_mass[body] == 0.0 or body_elastic_index[body] >= 0:
    diag = _mat66_identity(1.0e30)

```

The body stays in the sparse structure, so joints reaching it still assemble and still act on its neighbours, but its own six rows are pinned, the solve returns $\delta_e \approx 0$, and the delta is not applied. The port had the discard without the pin, which is what made $\mathbf{H}_{re} \delta_e$ large; with the pin the discarded delta is genuinely zero. This mirrors the local path, where `solve_rigid.body` has always returned early on elastic bodies.

3. The elastic solve is unified in both paths (`solver_vbd.py`). `solve_frame=True` at all three sparse launch sites. Frame and modes are solved together, once, from one contact evaluation, in both the local and the block-sparse path.

Verification

	before	frame excluded	unified (now)
indefinite blocks (of 320)	132	0 (trivially $\mathbf{I}_6$)	0
$\lambda_{\min}$	-6.5×10^8	+1.0	$+1.88 \times 10^5$
linear residual	0.47–1.25	10^{-7}	$\leq 1.2 \times 10^{-5}$
max Cauchy–Schwarz ratio	3.66	0	0.9962

The sparse and local paths now produce identical block statistics. The Cauchy–Schwarz ratio sits just inside the bound and never crosses it, which is what a genuine $\mathbf{J}^\top \mathbf{K} \mathbf{J}$ looks like when a contact couples the frame and the modes almost maximally. Over 90 frames both paths stay bounded and agree (base z : sparse $0.640 \rightarrow 0.436$, local $0.642 \rightarrow 0.187$). `test_vbd_sparse_articulation` passes 14/14.

Still open

- No guard in the tiled solve.** The `isfinite` check should be a residual check ($\|\mathbf{H}\delta + \mathbf{g}\| > \varepsilon \|\mathbf{g}\|$ falling back to Jacobi). It would have caught this on the first iteration of the first substep.
- No regression test.** A drop test cannot reproduce it: the deficiency is in the friction tangent plane and needs lateral slip. The cheapest real test asserts $\lambda_{\min} > 0$ on the assembled block.
- `test_reduced_elastic_body` reported 17 failures before this work; whether they predate it is unconfirmed.

Dead ends

Recorded so they are not re-run: stiff two-block Gauss–Seidel ($\rho < 1$ throughout, and a spectral radius cannot see a δ that does not solve $\mathbf{H}\delta = -\mathbf{g}$); stale force accumulators; the implicit frame/modal mass coupling; the neighbour solver (0.9892 vs. 0.9893); sign or index errors in the cross blocks (each is a correct derivative of *its own evaluation*); the minimal 2D model, too small to reproduce anything. Pinning the sparse block alone was tried early and still diverged: necessary, not sufficient.